Mahbub Alam, PhD

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About me

I hold a **PhD in Mathematics** with postdoctoral experience, specializing in probability theory, stochastic processes, and statistical modeling. I have developed Python-based machine learning models for **credit risk assessment, breast cancer prediction, and fruit classification**, using logistic regression and random forest and k-NN algorithms. Additionally, I have used SQL to extract, clean, and transform large datasets from the MIMIC-IV database. Looking forward to applying my analytical background, coding skills, and detail-oriented mindset in an industry environment as a Quant or Data Scientist.

Skills

Analytical and Mathematical skills

- Developed supervised ML models for assessing credit risk, predicting breast cancer and detecting fruits (regression and classification); experience in exploratory data analysis, visualization *(2 years)*
- Advanced expertise in Probability Theory, Stochastic Processes, Mathematical Modeling, and Classical Statistics (Hypothesis Testing, Bayesian Methods, Regression) *(10 years)*

Programming

- *Python*: pandas, Scikit-learn, Keras/TensorFlow, Matplotlib, NumPy, Statsmodels, SciPy; OOP; experience in GCP *(7 years)*
- *R, SQL (1 year)*, *Git, Shell (5 years)*; comfortable working in Linux-based environments

Soft skills

- Problem solving, Fast learner, Communication, Teamwork, Leadership, Growth Mindset
- Languages: English (fluent), Swedish (B2 – Uppsala University), Hindi (fluent)

Courses

- Applied Deep Learning, *Uppsala University*
- Machine Learning for Trading with GCP
- Machine Learning Specialization *(by Andrew Ng), Coursera*

Work Experience

Postdoctoral researcher, Uppsala University, Sweden — Jan 2022 – Dec 2024

- Stochastic analysis, modeling large datasets, central limit theorem, Brownian motion.
- Studied and implemented machine learning methods using Keras, Scikit-learn.
- Planned, organized and led projects.

PhD Candidate, TIFR, Mumbai, India — 2016 – 2021

- Conducted research on applications of statistics and probability in number theory.
- Performed numerical experiments using NumPy, Pandas, SciPy, Matplotlib.
- Published research in top journals, presented at international conferences.

Projects (GitHub link)

Selected projects demonstrating the application of analytics, machine learning and SQL in real-world data scenarios.

Machine Learning projects

- **BreastCancerPredictAI**, **Credit_Risk_Analysis**, **Fruit_Classifier_ML** : Implemented supervised learning models using Scikit-learn, Keras, including k-Nearest Neighbors (kNN), Linear Regression, Logistic Regression, Decision Trees, and Random Forests. (GitHub link)

Data modeling in data warehousing

- Designed and optimized complex SQL queries to extract, clean, and transform large-scale clinical data from the MIMIC-IV Clinical Database.

Data Analysis

- CarData EDA: Exploratory data analysis, data cleaning, data visualization, regression models (linear, polynomial, logistic regression) to study relationships between variables

Solving Advent of Code and LeetCode problems

- Sharpening algorithmic problem-solving skills through Advent of Code programming challenges.

Education

PhD in mathematics, TIFR Mumbai 2016 - 2021

- Statistical analysis of lattices, central limit theorems, Brownian motion

Master's in Mathematics, Indian Statistical Institute, Kolkata 2014 – 2016

- Advanced mathematics, Statistics

Personal

Planning to settle in Sweden with my partner, who works at the Department of Pharmacy at Uppsala University.